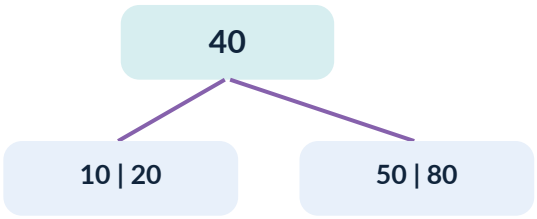
B-tree (balanced search tree index)

TREES, INDEXES & EVENT LOGS

Finding saved data and remembering changes are different operations.

SCAN / LEARN

THE SYSTEM AT A GLANCE



EVENT LOG

- 01 WRITE
- 02 COMMIT
- 03 REPLAY

FOUR IDEAS THAT MATTER

1 B-TREE (BALANCED SEARCH TREE)

A tree-based database index helps find exact or ranged values without examining every record.

2 INDEX IS NOT THE DATA

Like a book index: it points toward records. Indexes improve some reads but add work during writes.

3 APPEND-ONLY LOG

A sequence of events added at the end. Good for following what changed and replaying processing.

4 REPLICATION & ORDER

Several machines may copy logs. Order and delivery guarantees must be defined; “stored” is not always “fully committed”.

MEMORY HOOK

An index finds an item. A log remembers what happened.

DISCUSS WITHOUT COMPUTERS

Why can adding more indexes slow down database updates?

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